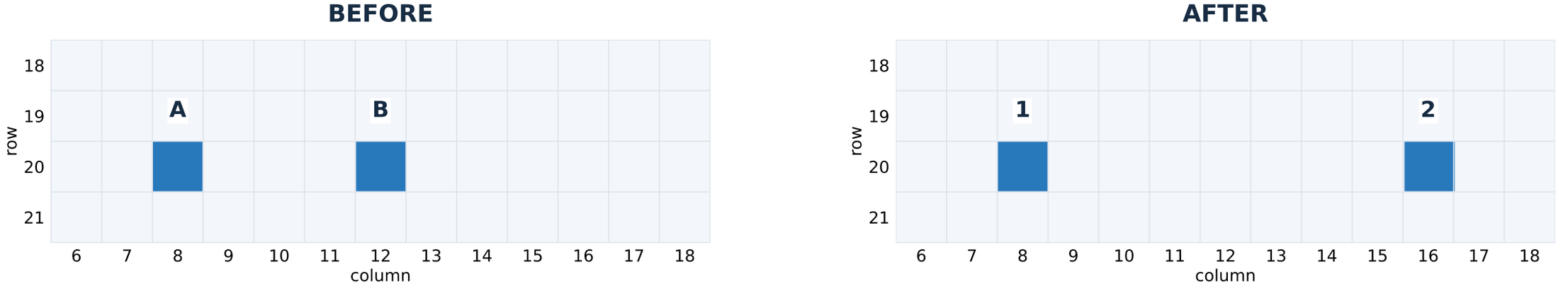
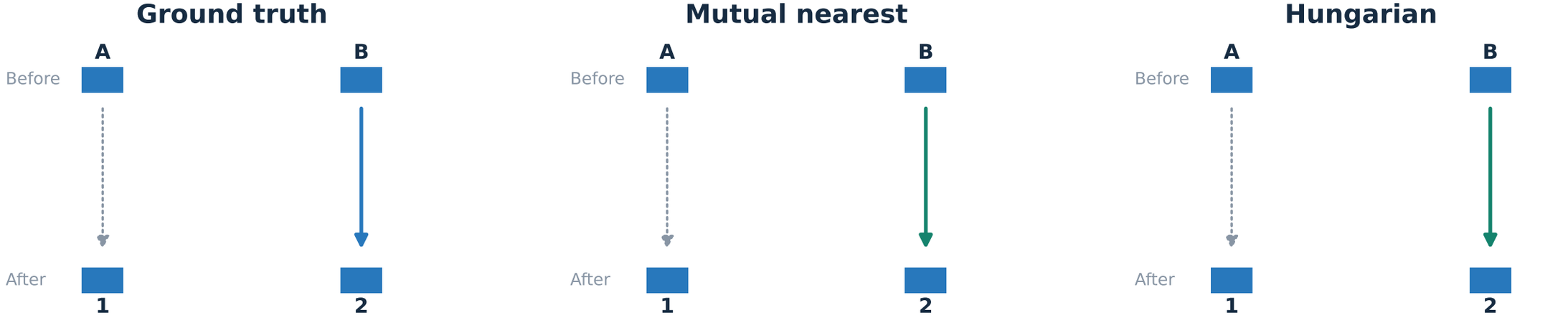
01 · One of two identical objects moves

Both methods match the moving object and omit the stationary object.



Frames are zoomed for display; row/column coordinates remain original.



Known fixture history

1 correct 0 wrong 0 missed

1 correct 0 wrong 0 missed

Solid green: correct **Solid red: wrong** **Dashed amber: missed** **Dotted gray: unchanged**

A stays at column 8. B moves from column 12 to 16. Removing unchanged A first prevents it from competing for B’s destination.

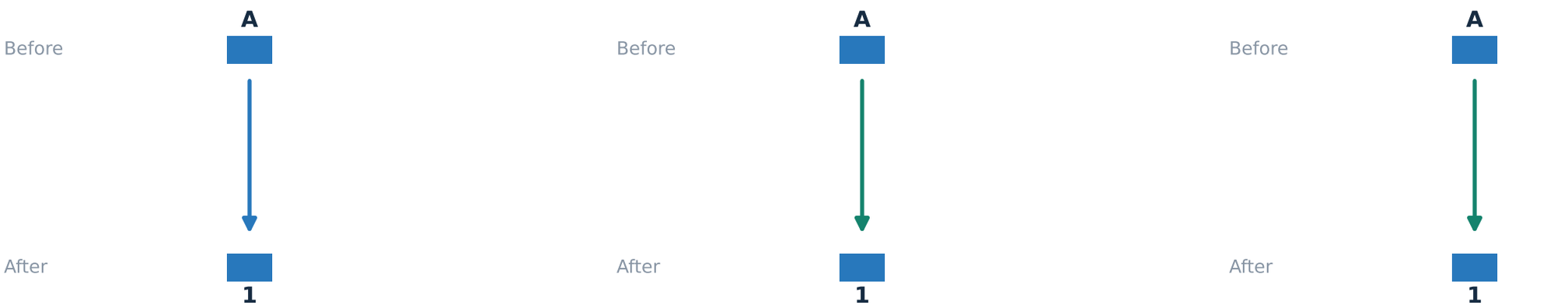
Recorded case: stationary_duplicate/0 · Fixture identity is used for scoring, never supplied to the matcher.

02 · A single object rotates 90° clockwise

Both methods get the pair and rotation right.



Frames are zoomed for display; row/column coordinates remain original.



Known fixture history

1 correct 0 wrong 0 missed

1 correct 0 wrong 0 missed

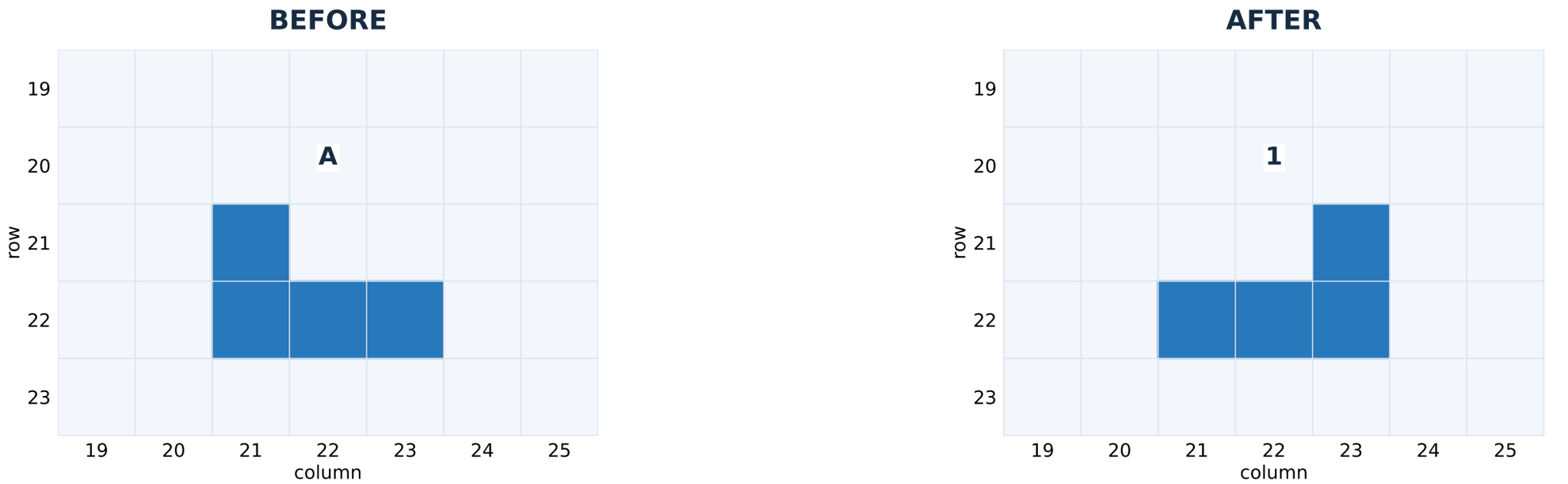
Solid green: correct **Solid red: wrong** **Dashed amber: missed** **Dotted gray: unchanged**

There is only one compatible foreground pair. Rotation is verified from the masks before assignment, so the choice of matcher makes no difference.

Recorded case: rotation_90/0 · Fixture identity is used for scoring, never supplied to the matcher.

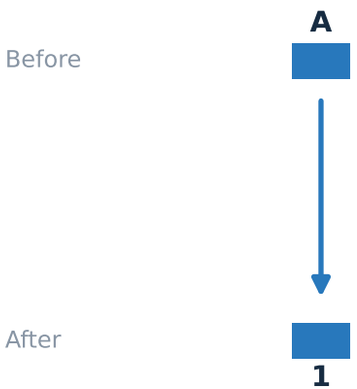
03 · A single object is reflected left–right

Both methods get the pair and reflection right.



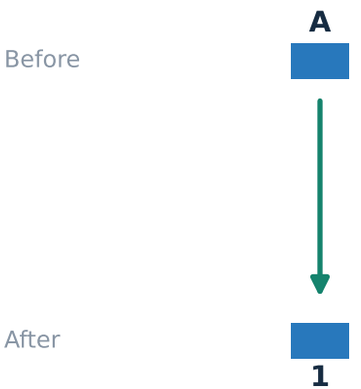
Frames are zoomed for display; row/column coordinates remain original.

Ground truth



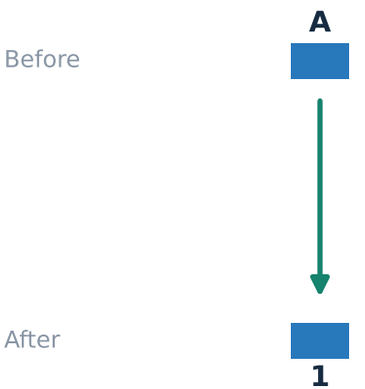
Known fixture history

Mutual nearest



1 correct 0 wrong 0 missed

Hungarian



1 correct 0 wrong 0 missed

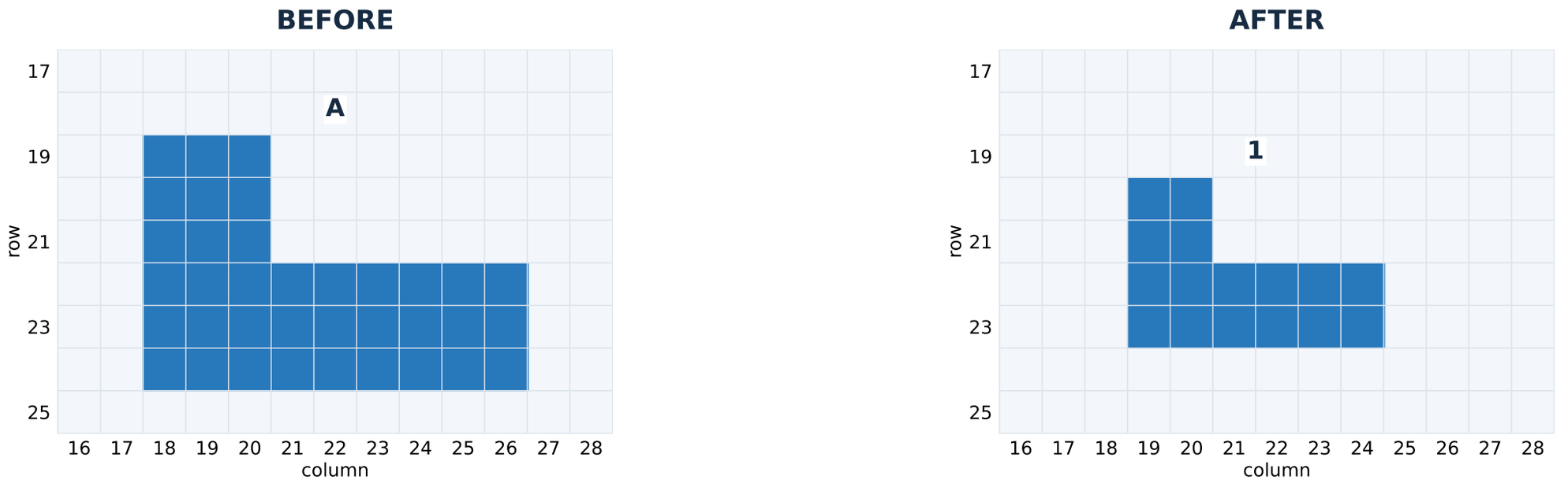
Solid green: correct **Solid red: wrong** **Dashed amber: missed** **Dotted gray: unchanged**

The chiral L changes handedness. A reflection across the vertical axis matches the masks; rotation alone does not.

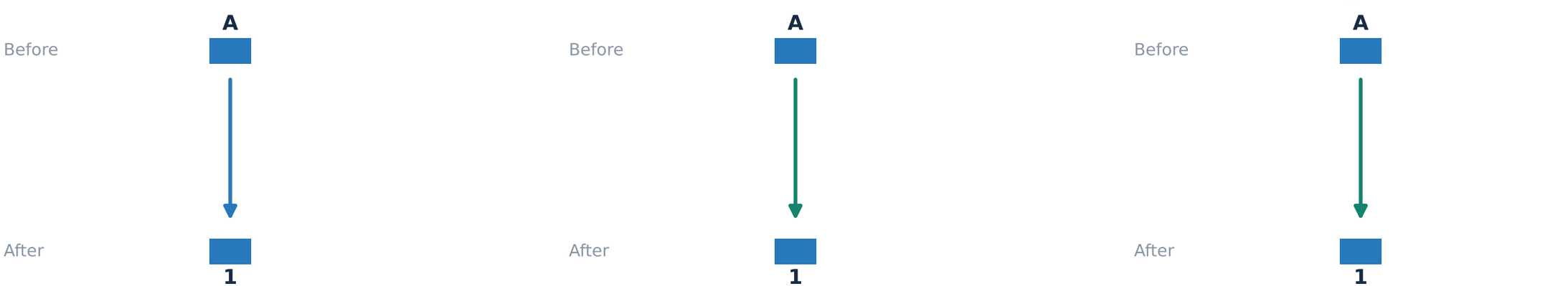
Recorded case: mirror_axis_|/0 · Fixture identity is used for scoring, never supplied to the matcher.

04 · An object shrinks from scale 3 to scale 2

Both methods get the pair and scale factor 2/3 right.



Frames are zoomed for display; row/column coordinates remain original.



Known fixture history

1 correct 0 wrong 0 missed

1 correct 0 wrong 0 missed

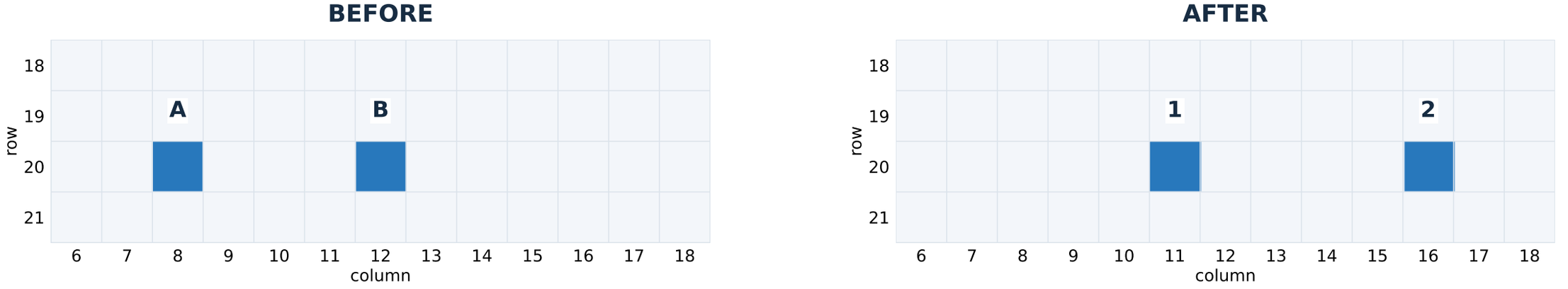
Solid green: correct **Solid red: wrong** **Dashed amber: missed** **Dotted gray: unchanged**

The colored pixels differ, but the oriented reduced masks are identical. The supported change is exact uniform scaling.

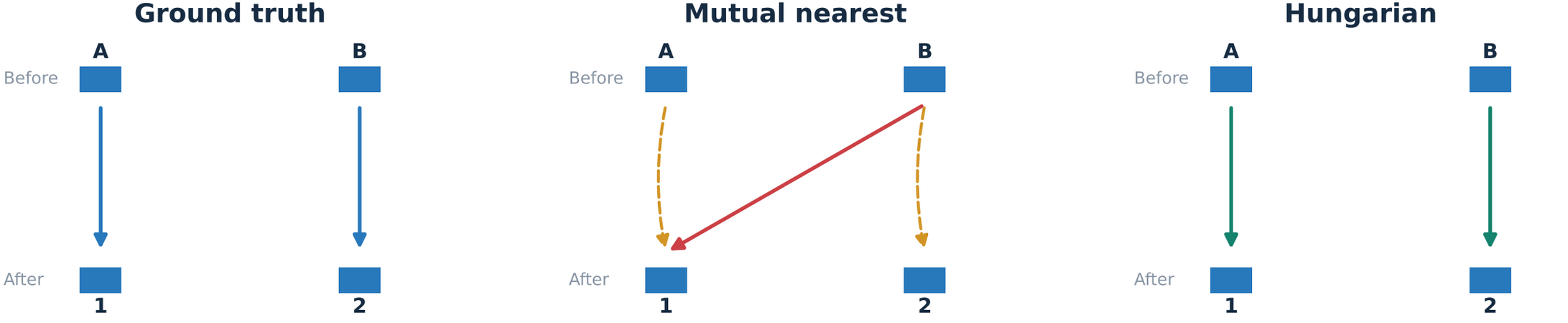
Recorded case: scale_3_to_2/0 · Fixture identity is used for scoring, never supplied to the matcher.

05 · Nearest locally, wrong globally

Hungarian gets both identities right. Mutual nearest chooses one wrong pair.



Frames are zoomed for display; row/column coordinates remain original.



Known fixture history

0 correct 1 wrong 2 missed

2 correct 0 wrong 0 missed

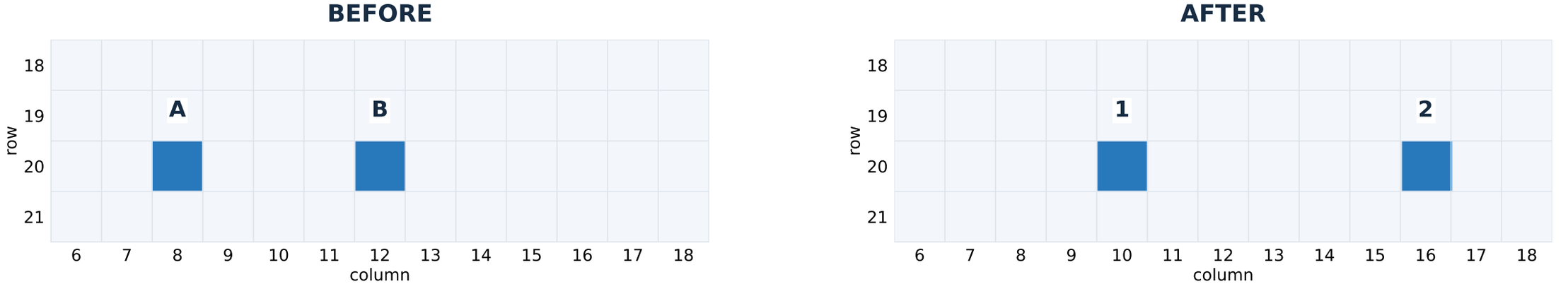
Solid green: correct **Solid red: wrong** **Dashed amber: missed** **Dotted gray: unchanged**

True moves: A 8→11 and B 12→16. Mutual nearest picks B→1 because its distance is 1. Hungarian picks A→1 and B→2, total distance 7 rather than 9.

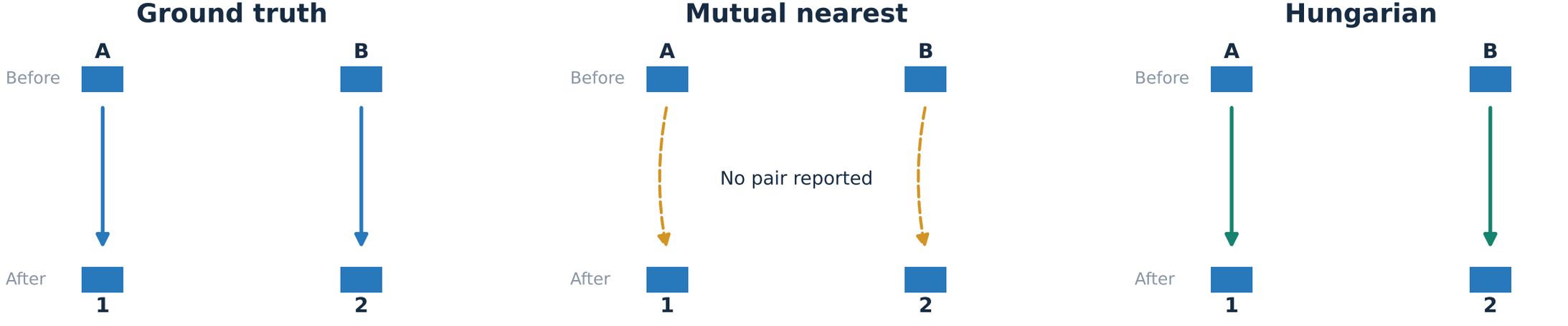
Recorded case: local_competition_ordered/0 · Fixture identity is used for scoring, never supplied to the matcher.

06 · A local tie hides a unique global assignment

Hungarian gets both right. Mutual nearest returns neither pair.



Frames are zoomed for display; row/column coordinates remain original.



Known fixture history

0 correct 0 wrong 2 missed

2 correct 0 wrong 0 missed

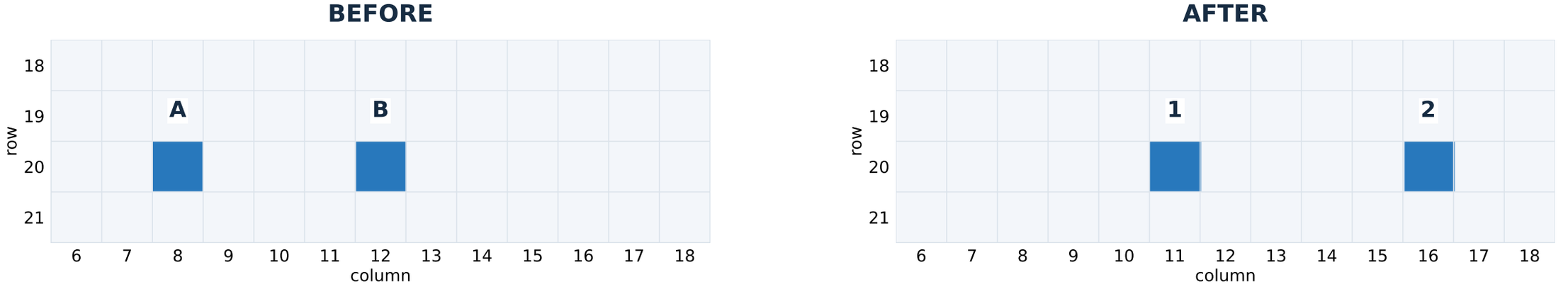
Solid green: correct **Solid red: wrong** **Dashed amber: missed** **Dotted gray: unchanged**

Before columns are 8 and 12; after columns are 10 and 16. Node 1 has two equally close predecessors. The complete minimum-cost assignment is nevertheless unique.

Recorded case: local_tie_global_unique/0 · Fixture identity is used for scoring, never supplied to the matcher.

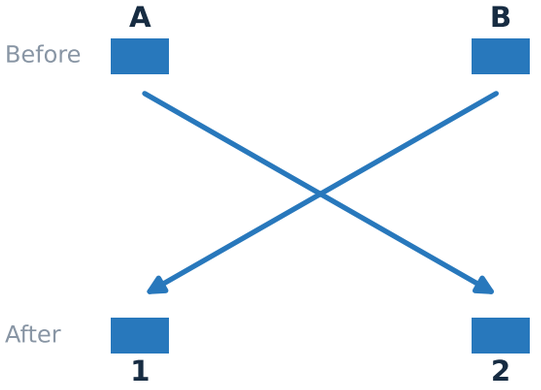
07 · Same images as case 5, different true history

Mutual nearest gets one right. Hungarian gets both identities wrong.



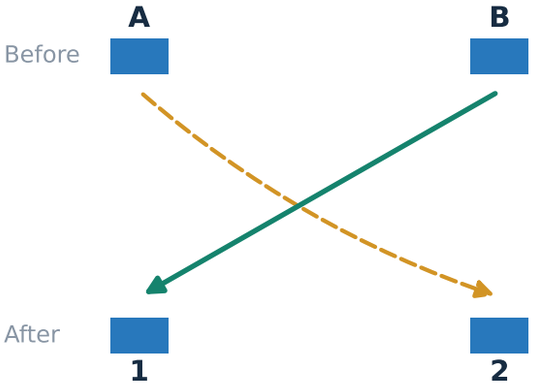
Frames are zoomed for display; row/column coordinates remain original.

Ground truth



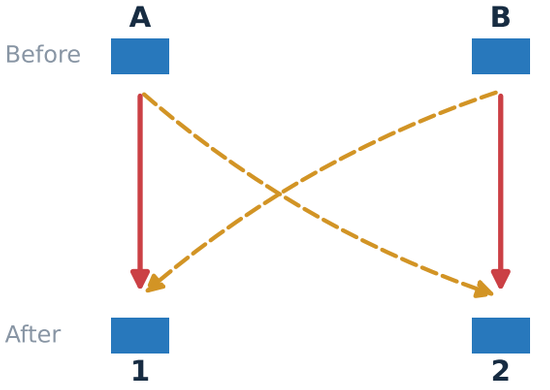
Known fixture history

Mutual nearest



1 correct 0 wrong 1 missed

Hungarian



0 correct 2 wrong 2 missed

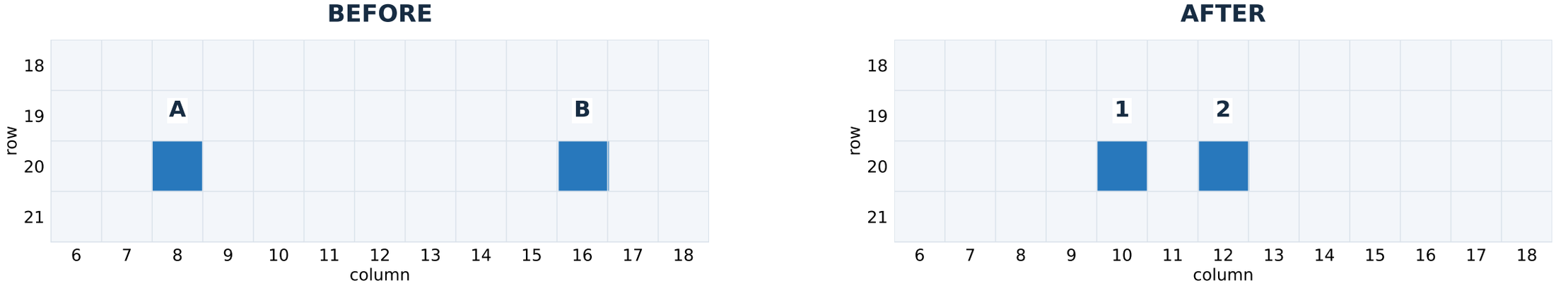
Solid green: correct **Solid red: wrong** **Dashed amber: missed** **Dotted gray: unchanged**

The actual moves are A 8→16 and B 12→11. The frames are identical to case 5: only the hidden fixture history differs. Our B→1 pair is now correct.

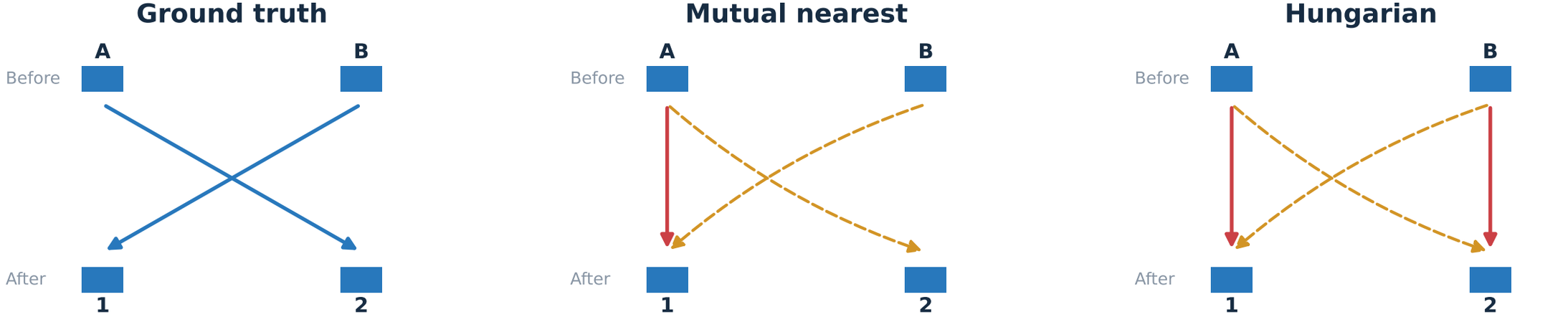
Recorded case: local_competition_crossed/0 · Fixture identity is used for scoring, never supplied to the matcher.

08 · Objects cross: shorter paths give wrong identities

Both methods make wrong matches; Hungarian makes two.



Frames are zoomed for display; row/column coordinates remain original.



Known fixture history

0 correct 1 wrong 2 missed

0 correct 2 wrong 2 missed

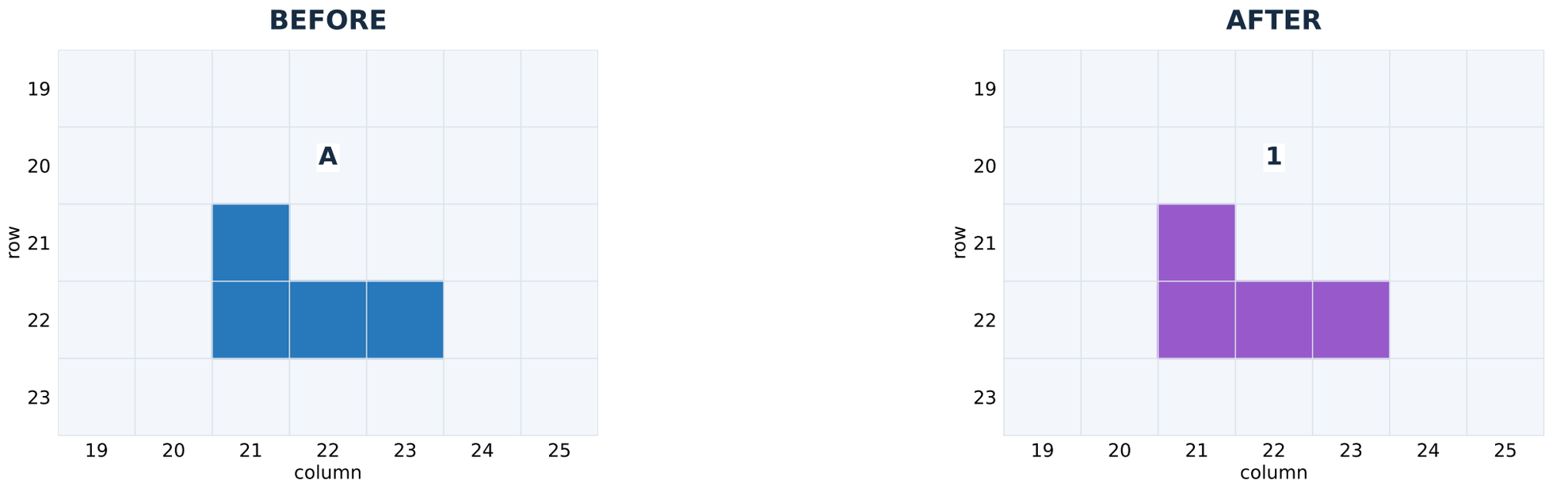
Solid green: correct **Solid red: wrong** **Dashed amber: missed** **Dotted gray: unchanged**

True moves: A 8→12 and B 16→10. The distance heuristic instead favors A→1 at column 10. Hungarian also pairs B→2 at column 12.

Recorded case: crossing/0 · Fixture identity is used for scoring, never supplied to the matcher.

09 · Replacement looks exactly like recoloring

Both methods incorrectly claim that the old object was recolored.



Frames are zoomed for display; row/column coordinates remain original.

Ground truth

Before: **A**

Old object disappears.
New object appears.

After: **1**

Known fixture history

Mutual nearest

Before: **A**

After: **1**

0 correct 1 wrong 0 missed

Hungarian

Before: **A**

After: **1**

0 correct 1 wrong 0 missed

Solid green: correct **Solid red: wrong** **Dashed amber: missed** **Dotted gray: unchanged**

The fixture deletes the blue object and creates a different purple object on the same cells. The images are indistinguishable from recoloring one persistent object.

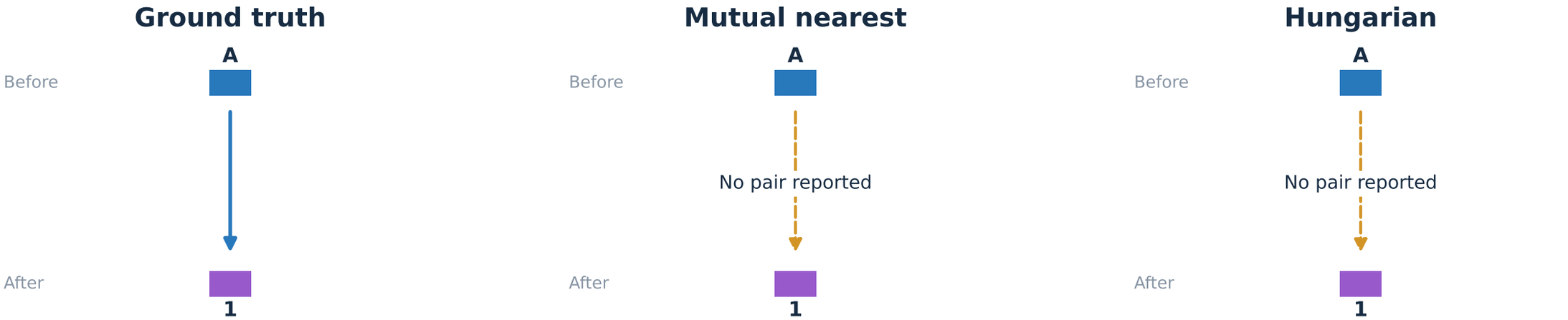
Recorded case: unrelated_changed_color/0 · Fixture identity is used for scoring, never supplied to the matcher.

10 · Rotation and recoloring happen together

Both methods leave the true pair unmatched, as the current rules require.



Target zoom. Two unchanged distractors are outside this view; both methods received the full frames.



Known fixture history

0 correct 0 wrong 1 missed

0 correct 0 wrong 1 missed

Solid green: correct **Solid red: wrong** **Dashed amber: missed** **Dotted gray: unchanged**

A both rotates and changes color. There is no combined-change bucket, so the true pair is reported as disappearance/appearance. This miss is a policy limit, not an assignment failure.

Recorded case: rotate+recolor/0 · Fixture identity is used for scoring, never supplied to the matcher.